Sorting Mechanism:

1. Introduction & Components:

The sorting mechanism is designed to automatically detect and separate balls based on their color (black and white). It plays an important role in the system by directing each ball to the correct path after the shooting stage. The mechanism uses an IR sensor to identify the color of the ball and a servo motor to control the direction of movement. The design focuses on simplicity, reliability, and fast response.

Component Name	Function	Image
Servo Motor (sg90)	Rotates the arm to direct the ball left or right	
IR Sensor (TCRT5000)	Detects ball color using reflected infrared light	
Ultrasonic Sensor	Measures distance using ultrasonic waves to determine if a ball is present in the sorting area.	
Arduino Uno	Controls the system and processes signals	

2. Actuator Sizing:

$T_t = T_{acc} + T_{mass}$

Density(PLA) = 1.240 g/cm^3 , m= 0.017 kg , infill = 20%

- T_{mass}

T=m*g*r

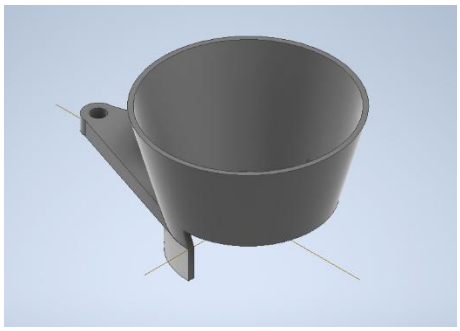
density = 1.240 g/cm^3 , m= 0.017 kg , infill = 20%

m= 0.017 * 0.2 = 0.0034 kg , g=9.81 m/s^2

Center of Gravity ➔ X= -11.800 mm , Y = 12.400 mm , Z=0

Center of motor ➔ X=-55.272 mm , Y =0 , Z=0

$r = -11.8 - (-55.272) = 43.472\ mm$



$Tmass = 0.0034 * 9.81 * 4.3472 = 0.145\ N.cm$

- T_{acc}

$\boldsymbol{T = I * \alpha}$

$\boldsymbol{I = I_{yy} + m\,r^2}$, I_{yy}= 14.871 kg mm^2

$\boldsymbol{I = 0.14871 + 0.0034 * 4.3472^2 = 0.213\ kg.cm^2}$

$\theta = \frac{\pi}{2} rad \quad , \quad \omega_\circ = 0, \quad t = 0.5\ sec$

$$\boldsymbol{\theta = \omega_\circ t + \frac{1}{2}\alpha t^2}$$

$\boldsymbol{\theta = 0 + \frac{1}{2} * \alpha * 0.5^2 = \frac{\pi}{2}}$ ➔ $\boldsymbol{\alpha = 12.56\ rad/s^2}$

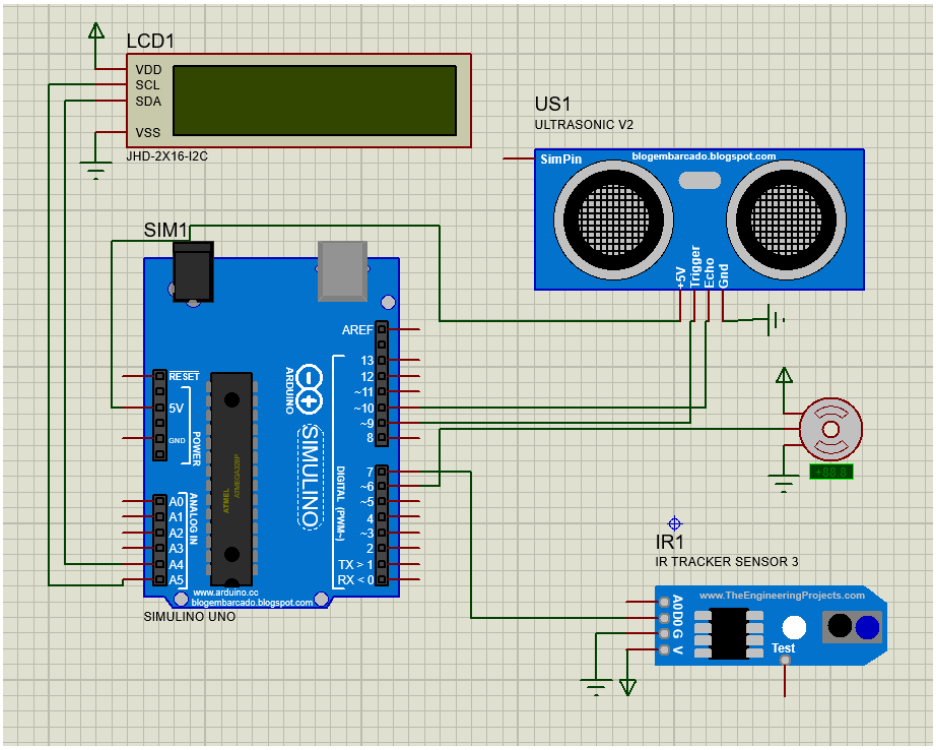
$\boldsymbol{T = 0.213 * 12.56 = 2.67\ N.cm}$

$T_t = T_{acc} + T_{mass}$ ➔ T_t = 2.67 + 0.145 = 2.815 N.cm

Safety Factor of (3)

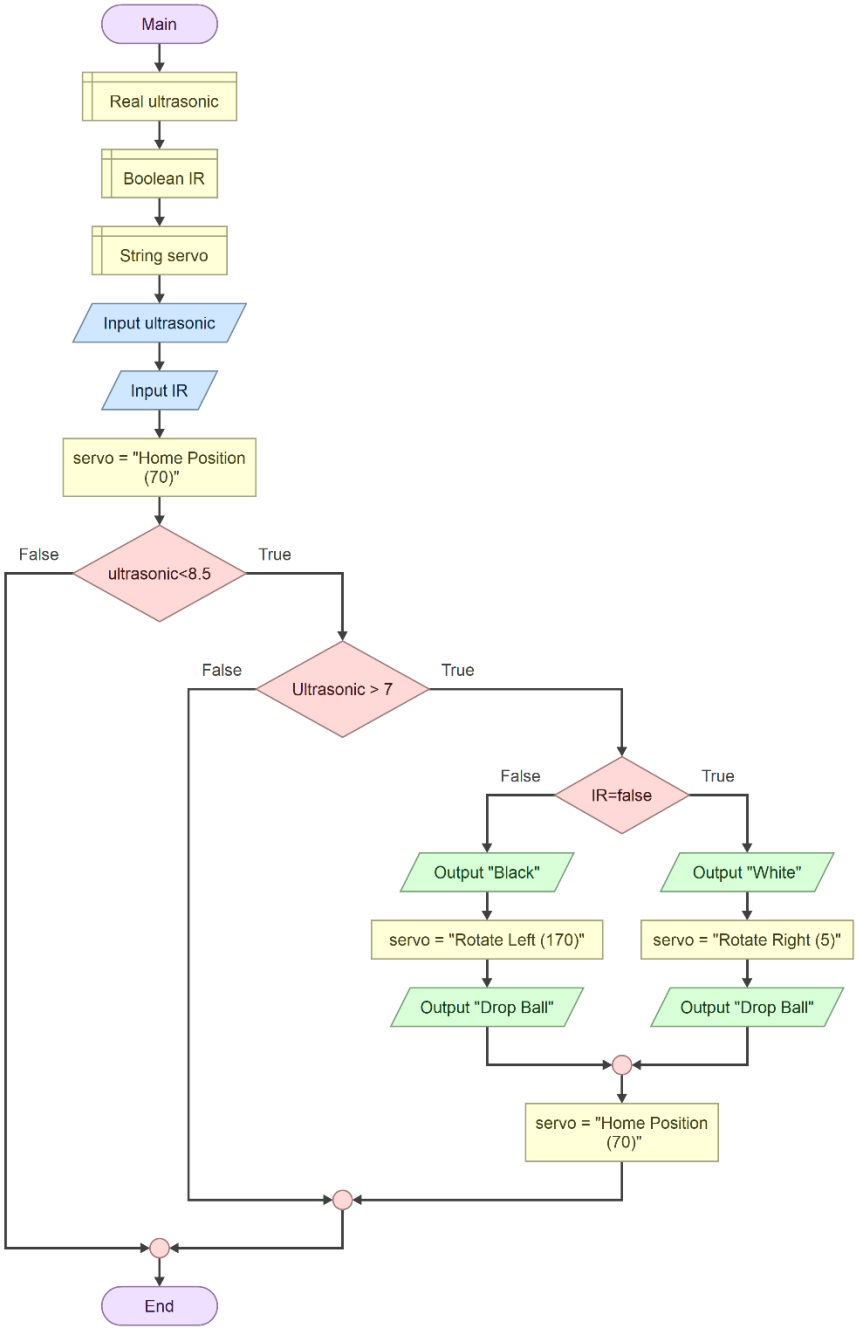
T_t=2.815 * 3= 8.445 N.cm **sg90 suitable**

3. Sorting Wiring:



4. Sorting code

```
#include <Servo.h>
#include <LiquidCrystal_I2C.h>

Servo myServo;
LiquidCrystal_I2C lcd(0x27, 16, 2);

int trigPin = 9;
int echoPin = 10;
int irPin = 7;
long duration;
float distance;
int choose; //the kid choose in app
int score=0;

void setup() {
  Serial.begin(9600);

  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(irPin, INPUT);
  myServo.attach(6);
  myServo.write(70);

  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Welcome to Element Explorer");
}

void loop() {

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH, 30000);


  distance = duration * 0.034 / 2;

  Serial.print("Distance: ");
  Serial.println(distance);

  if (distance > 7 && distance < 8.5) {
  delay(1000);

    int irState = digitalRead(irPin);
    Serial.print("IR: ");
    Serial.println(irState);

    if (irState == LOW) {
      myServo.write(5);
    } else {
```

```
      myServo.write(170);
    }
    delay(1000);
    myServo.write(70);

    lcd.clear();
    if(choose == irstate){
      lcd.setCursor(0,0);
      lcd.print("Great Job!");
      score++;
      lcd.setCursor(0,1);
      lcd.print(score);
    }
    else{
      lcd.setCursor(0,0);
      lcd.print("Try again ");
      lcd.setCursor(0,1);
      lcd.print(score);
    }
  }
}
```

5. Sorting Flow Chart: